

Project Title: Content-Based Epidemic Intelligence for Recurring Seasonal Epidemics**Problem Statement:**

Infectious diseases can still spread rapidly, causing many people to become sick within a short period of time. Therefore, early detection of increasing disease activity and timely notification are important for supporting public health response and helping reduce further transmission.

However, official disease surveillance systems recognize outbreaks only after cases have been formally confirmed and reported, which can delay the recognition of emerging or intensifying disease activity. This limitation creates a need for public health agencies and surveillance teams to obtain earlier complementary signals of increasing seasonal epidemic activity.

Informal-source surveillance may provide an earlier source of information. Informal disease reports can contain epidemiologically relevant information, such as case counts, severity, and symptoms. However, existing approaches that rely on how often reports about a particular disease appear within a period of time may not reliably reflect actual epidemiological activity. For example, Ganser et al. (2022) found that HealthMap/EIOS detected only 22 of 238 seasonal influenza outbreaks within a two-week window when compared with WHO FluNet data across 24 countries over seven years of monitoring, the exact comparison this project builds on.

Even modest improvements in detection lead time could provide public health agencies and surveillance teams with additional time to support resource allocation and response during recurring seasonal epidemics. Beyond this immediate application, a validated content-based approach could also provide a foundation for future research on newly emerging diseases, where early information may first appear in informal reports before sufficient structured surveillance data become available.

Objective: Develop and evaluate a content-based epidemic detection system for seasonal influenza, to determine whether analyzing the content of informal disease reports can detect seasonal epidemics earlier than a frequency-count baseline.

Data Sources: The project uses two genuinely heterogeneous data types: structured numeric surveillance and unstructured multilingual text. Both structured and unstructured sources were confirmed live end to end before this proposal was drafted, not assumed, as summarized below.

Data	Structure/Data Type	Source	Purpose	Data Accessibility	Challenges
Influenza , covering 6 countries; 2015-2024	Type: Structured/Numeric Data: One row per country per epidemiological week Primary Field: Specimens Processed and Influenza-positive Specimens, used to calculate weekly percent positivity	WHO FluNet	Primary ground truth to decide when the influenza epidemic is considered to have started	Publicly accessible through WHO's REST API. Data can be requested in CSV format and filtered by country and year.	Preprocessing/Cleaning: Calculate weekly percent positivity and identify onset when positivity is $\geq 10\%$ for two consecutive weeks.
Informal Disease Reports	Type: Unstructured/Text Data: Multilingual news, with a new raw file every 15 minutes Primary Field: V2Locations field, identifying real-world locations mentioned in each article and identifying article associated with our 6 study countries	GDELT GKG 2.0 raw files	Provides flu-related informal reports for extracting epidemiological information and generating the content-based detection signal.	Publicly accessible through GDELT GKG 2.0 raw files without authentication. Full text is retrieved from the source URLs of matched news articles. *Some article URLs lack retrievable full text (60–90% usable by country), retrievability is tracked by country, and unretrievable articles are logged and excluded from analysis.	Preprocessing/Cleaning: 1. A simple search for “flu” also matches unrelated words such as “fluid” or “influencer.” A word-boundary rule is used to eliminate false positives. 2. GDELT's automatic location tagging may assign articles to the wrong country. GDELT location tags are cross-checked against locations extracted from article text, and mismatches are excluded.

Methodology: The proposed pipeline runs in five stages:

1. **Data ingestion and filtering:** Raw GDELT GKG 2.0 files are downloaded and parsed for the study's country-season windows. Articles are filtered for flu-related terms using complete-word matching to avoid substring false positives and are restricted to the six study countries using GDELT's V2Locations field.
2. **LLM-based few-shot event extraction:** Filtered articles are processed using few-shot LLM extraction and converted into a structured JSON schema containing disease signal, severity, symptoms, strain/serotype, location, and confidence. This approach is informed by Health Sentinel (Pant et al. 2025), which reported that LLM-based few-shot extraction outperformed traditional NER and QA-based methods on the same event-extraction task (event-level F1: 0.68 vs. 0.40).
3. **Rule-based relevance filtering:** Extracted events are retained only when they contain a concrete case, severity, or symptom signal beyond a general disease mention, excluding general health-information articles (e.g., “what is the flu” explainer

pieces) that do not reflect an actual reported event. A fixed, auditable rule set is used rather than a learned classifier so that each filtering decision can be traced to a specific rule.

4. **Cross-document linking:** Multiple reports are aggregated into a single weekly signal using a deterministic join by country x epidemiological week. If this approach proves too coarse, such as by combining unrelated events reported within the same week, an embedding-similarity clustering approach will be evaluated as an alternative. The method producing cleaner weekly signals on the calibration subset will be selected.
5. **Threshold-based change-point detection:** Change-point detection is applied identically to the content-based and frequency-count signals to identify the first sustained increase above the early-season baseline. For each country-season, the detection point is defined as the first week in which the signal exceeds its early-season baseline mean by more than 2 standard deviations and remains elevated across consecutive weeks. This differs from the FluNet ground-truth onset rule, which uses a fixed 10% positivity threshold. The same detection rule, applied to both methods, ensures a fair comparison.

Evaluation: FluNet onset dates serve as the ground-truth reference and are established before content analysis. Approximately 55 country-seasons are split into a calibration set, used to tune extraction prompts and detection thresholds, and a held-out test set, kept untouched until the final comparison. For each country-season, detection lead time is measured as the number of days before or after the FluNet onset date that each method first detects a signal. Lead time is calculated identically for the content-based method and the frequency-count baseline and compared across the held-out set.

Validation: The head-to-head lead-time comparison on the held-out test set serves as the validation method. Unlike traditional machine-learning validation, which often uses cross-validation to compare predictions against known categories or values, this project evaluates detection timing rather than category prediction. Therefore, comparing the content-based and frequency-count lead times on data that were not used for tuning provides validation for the detection-timing task.

Project Management: The project is divided into four phases, as outlined below. The schedule front-loads the core influenza pipeline and evaluation, leaving the remaining time for refinement and follow-up extensions.

Phase	Task	Dates	Status Check-ins
Data Prepa-ration	Finalize ground-truth reference data and onset dates; resolve extraction infrastructure; ingest GDELT data at scale for the six study countries.	09/15-09/28	09/18: Ground truth & infrastructure completed 09/25: ~80% of GDELT data ingested 09/28: GDELT ingestion completed
Model Develop-ment	Build and run the extraction pipeline, relevance classification, and cross-document linking, followed by change-point detection on both the content-based and frequency-count signals	09/29-10/19	10/02: Initial extraction pipeline operational. 10/09: LLM extraction tested and refined on sample articles 10/16: Validation sample output produced
Evaluati-on & Analysis	Compute detection lead times, refine thresholds using the calibration set, and begin extending the approach to a secondary track (e.g., dengue) as time permits.	10/20-11/10	10/27: Detection thresholds calibrated 11/06: Lead times computed for both methods
Report	Finalize country-level relative-intensity mapping and the writeup, and prepare the final report and presentation.	11/11-12/01	11/27: Final analysis, report, and presentation completed.

Risk Identification, Monitoring, and Mitigation:

Risk	Potential Impact	Monitoring / Metric	Mitigation
Uneven GDELT coverage across countries	Insufficient informal-report data may reduce detection reliability in poorly covered countries.	Per-country retrievability and content-depth rates.	Monitor country-level data availability, prioritize adequately covered countries, and document thin coverage as a limitation.
LLM extraction accuracy	Incorrect extraction of epidemiological information could affect the content-based signal.	Extraction accuracy on a held-out labeled validation sample.	Evaluate on a held-out labeled sample, refine few-shot prompts, and report extraction accuracy.
Detection-thres-hold sensitivity	Detection timing may vary across country-seasons with different baseline signal levels.	Detection threshold and baseline mean/standard deviation by country-season.	Calibrate each country-season's threshold using its own baseline mean and standard deviation rather than applying one universal threshold.

References

1. Ganser, I., Thiébaud, R., & Buckeridge, D. L. (2022). Global Variations in Event-Based Surveillance for Disease Outbreak Detection: Time Series Analysis. *JMIR Public Health and Surveillance*, 8(10), e36211. <https://doi.org/10.2196/36211>
2. Pant, D., Grandhe, R. R., Agrawal, J., Kalra, J. S., Kumar, S., Khanna, S., Samaria, V., Paul, M., Khalikar, S. V., Garg, V., Chauhan, H., Verma, P., VSSG, A., Khandelwal, N., Dhavala, S. S., & Mathew, M. (2025). Health Sentinel: An AI Pipeline For Real-time Disease Outbreak Detection. *arXiv preprint*, <https://arxiv.org/abs/2506.1954>